

The earliest evidence of single-celled fossil life exists in rocks that are 3.5 billion years old.

Billions of years ago, algae produced much of the oxygen in the air today.



Megacerops

Tiny ancestors evolved into large mammals that replaced the dinosaurs.

23 MYA

Neogene Period

Many familiar groups of mammals, such as rodents, primates, antelopes, and cats, evolved in the Neogene, while flying birds diversified in the skies.



Thylacosmilus

Predatory mammals, including saber-toothed cats, hunted grazers on the open grassland.

2.6 MYA



Quaternary Period

Mammals and birds survived Quaternary ice ages, but one species—humans—drove many others to extinction by hunting and habitat destruction, causing the biggest mass extinction of modern times.

Red lines indicate mass extinctions.

201 MYA

Jurassic Period

The peak of the Age of Dinosaurs saw the evolution of giant reptiles on land and in the seas, which included the largest land animals of all time.

TRIASSIC-JURASSIC EXTINCTION



Asteroid impact

A space rock colliding with Earth wiped out a quarter of all species.



Eoraptor

The first dinosaurs, some two-legged, evolved from small reptiles that survived the Permian-Triassic extinction.

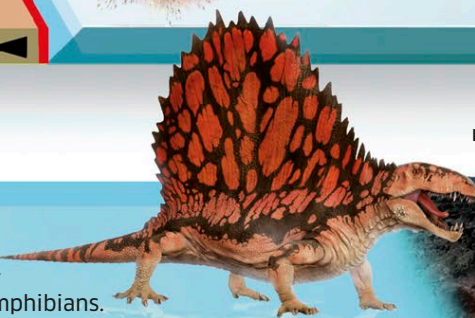
Periods of time

Earth's prehistory is marked by a timescale divided up into geological periods. Each period represents a length of time that has left its mark in rocks with fossils and other evidence.

299 MYA

Permian Period

Continents dried up to favor scaly reptiles over amphibians. The period ended with violent eruptions causing the biggest of all mass extinctions.



Dimetrodon

As moist-skinned amphibians declined in a drier world, reptiles such as this carnivore took over.

PERMIAN-TRIASSIC EXTINCTION



Triassic Period

New kinds of forests containing conifers and cycads were inhabited by the first dinosaurs until a possible asteroid impact brought about mass extinction.

251 MYA



Elginerpeton

Fishes with strong, fleshy fins—such as Elginerpeton—paved the way for the evolution of walking limbs.

Devonian Period

More kinds of fishes evolved in the oceans, while vertebrate animals and trees appeared on land. Climate change near the end of the Devonian Period caused mass extinction.

LATE DEVONIAN EXTINCTION

375-360 MYA

Ordovician Period

The oceans of the Ordovician teemed with invertebrates and primitive fishes, but ended with an ice age that drained shallow seas and caused a mass extinction.



Orthoceras

This relative of squid had grasping arms and a body inside a long, cone-shaped shell.

416 MYA

433 MYA

Climatius

This fish had a sharklike skeleton of rubbery cartilage, but bony, spiny fins.

Silurian Period

Jawed fishes and coral reefs appeared for the first time, while distant relatives of spiders and centipedes started to crawl onto land.

